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CAP V-SAR SIMULATION SYSTEM

Flight Operations Handbook & Multi-Station Training Manual

Standardized procedures for Civil Air Patrol electronic search and rescue training, featuring the Cessna 182T NAV III Mission Observer station, N499CP flight planning, and distributed exercise simulation.

Document ID: CAP-VSAR-FOH-2026-01

Revision: 1.3 (Training Progression: SAREX Rehearsal)

Target Airframe: Cessna 182T NAV III / G1000

Simulator Target: X-Plane 12 & Air Manager 4.2+

Exercise Host: ms01:8700 (Composite Stack)

Date: September 2026

1. System Overview & Operational Architecture

The Civil Air Patrol Virtual Search & Rescue (CAP V-SAR) training simulator provides end-to-end, procedurally authentic training for CAP aircrews, Mission Observers (MO), Mission Radio

Operators (MRO), and Incident Command staff. The architecture separates simulation physics from instrument display and exercise coordination.

1.1 Distributed Multi-Station Architecture

Unlike single-seat flight simulators where one user controls all aircraft systems, CAP V-SAR implements a true crew-resource-management (CRM) environment partitioned across distinct functional layers:

- **Flight Deck / Simulator (X-Plane 12):** Runs aircraft aerodynamics and the custom `PI_capvsar_df.py` Python plugin, probing real terrain for line-of-sight propagation, cone of confusion, and 406 MHz burst cadences.
- **Mission Observer (MO) Station:** Runs on a dedicated secondary monitor or laptop via Air Manager / Air Player, presenting the G1000 MFD, GMA 1347 Audio Panel, Direction Finder head, and Technisonic TDFM-136B radio.
- **Mission Planning Module (N499CP Flight Planner):** Runs as a containerized service on `ms01:8090`, generating geodetic search patterns and pushing flight plans to the simulator and exercise server.
- **Exercise Control & Mission Base (Web Console):** Hosted on `ms01:8700`, providing real-time radar positions, scenario fault injection, message logging, and debrief track analysis.

OPERATIONAL NOTE — VOICE COMMUNICATIONS Inter-crew and mission base communications are routed via an authentic radio emulation layer backed by a dedicated Mumble server (port 64738 on ms01). The MO transmits on CAP FM frequencies through the TDFM-136B and GMA 1347 COM3 channel.

2. Flight Planning & Search Pattern Module

Pre-flight mission planning and in-flight search pattern execution are powered by the **N499CP Flight Planner** (deployed on `ms01:8090`). The planner combines FAA sectional raster charts with real-time performance polar calculations for the Cessna 182T / 172S NAV III.

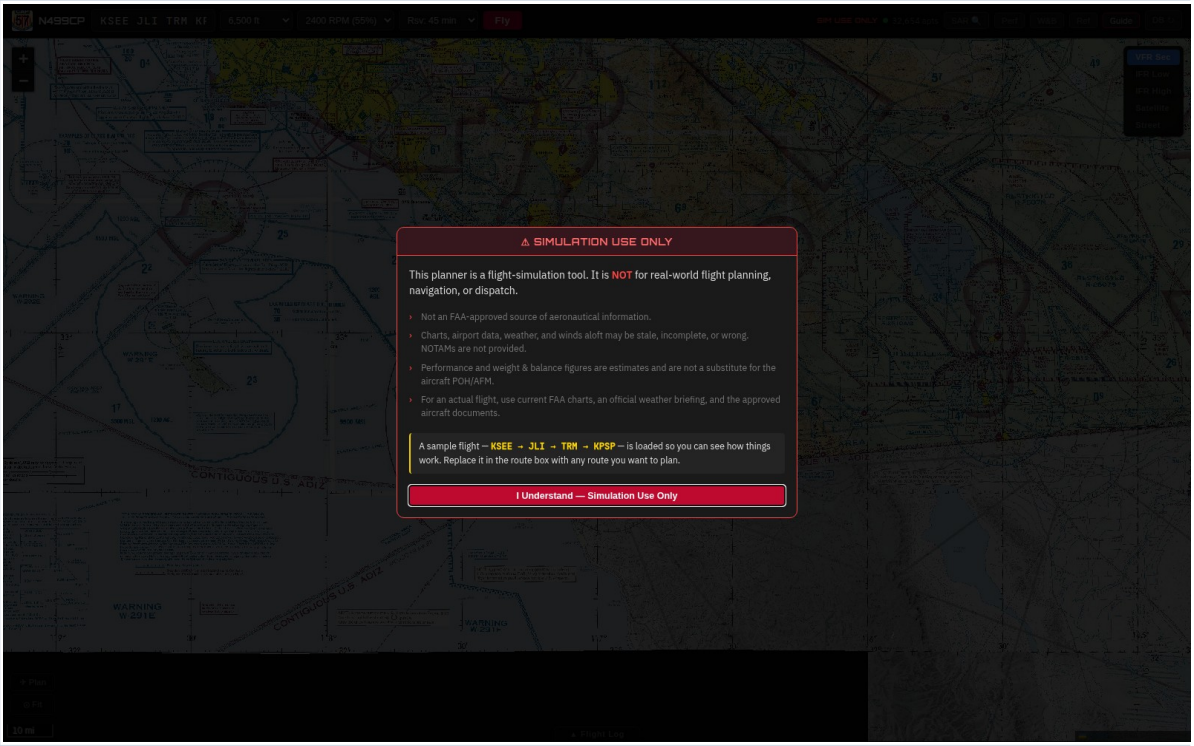


Figure 1: N499CP Flight Planner Moving Map with FAA Sectional Overlay & Nav-Log Drawer

2.1 CAP Search Pattern Generators

Clicking the **SAR** button on the main navigation toolbar launches the geodetic pattern generation tool. Three standardized CAP search patterns are supported:

Expanding Square (SS)

Used when searching for an Emergency Locator Transmitter (ELT) or missing aircraft with a known Last Known Position (LKP) or distress alert datum. The pattern expands outward in concentric squares:

- **Leg Lengths:** Successive legs follow the progression *1×S, 1×S, 2×S, 2×S, 3×S, 3×S...* where *S* is track spacing.
- **Track Spacing:** Configurable between 0.5 NM and 2.0 NM depending on terrain, visibility, and target size.
- **Turns:** Selectable Right (standard) or Left turns at precise 90° cardinal angles.

Creeping Line (CS)

Used to sweep an elongated search corridor, highway, valley floor, or river drainage where the

wreckage is suspected along a route axis:

- Flies cross-track sweeps perpendicular to the advance axis.
- Sweeps alternate left and right across the track corridor (default width 3.0 NM).
- Advances along the route by track spacing *S* after each cross sweep.

Parallel Track (PS)

Used for large-area systematic searches across CAP quadrangles. The aircraft sweeps parallel tracks back and forth across a rectangular search box defined by corner coordinates, axis heading, length, and width.

2.2 Simulator & Console Export Bridges

Once generated, the search pattern provides two immediate export capabilities:

1. **Export X-Plane 12 (.fms):** Generates a standard FMS 1100 format flight plan containing departure, enroute fixes, and all generated search pattern waypoints (e.g. `SS01`, `SS02`...). The pilot can load this directly into the G1000 MFD flight plan catalog.
2. **Push to V-SAR Console:** Transmits the planned search corridor over LAN to the exercise console (`http://exercise:8700`). The server creates a live scenario entry and, if an exercise sortie is active, broadcasts a tasking message to Mission Base.

FLIGHT TIP — FUEL RESERVES & OPS NORMAL The nav-log drawer computes true airspeed, ground speed, and fuel burn for the search pattern based on current GFS winds aloft. Always ensure estimated fuel allows for the mandatory CAP 1-hour daytime VFR reserve before committing to deep valley creeping-line searches.

3. Mission Observer (MO) Cockpit & Avionics Guide

The Mission Observer occupies the right cockpit seat of the Cessna 182T NAV III. The dedicated MO panel (Air Manager UUID `a4c8e710-5646-49e1-931f-4aeeb51e4490`) consolidates the entire mission observer workstation onto a single 1920×1080 canvas.



Figure 2: Dedicated C182T MO Station — Layout 1 (Becker SAR-DF 517 & TDFM-136B)

3.1 Instrument Arrangement & Functional Roles

Garmin G1000 MFD	Overlay (0b852ee1...)	Multi-Function Display showing synthetic vision, moving map, terrain alerts, and flight plan waypoints. Features transparent active screen cutout for popouts or video streaming.
MO Audio Panel	Garmin GMA 1347 (1337420b...)	Copilot/Observer audio control. Manages headphone volume, ICS intercom, and separate COM1 / COM2 / COM3 receiver and microphone routing.
Direction Finder	Becker 517 / RT-600	Precision multi-band electronic homing receiver for 121.5 MHz VHF, 243.0 MHz military UHF, and 406.025 MHz COSPAS/SARSAT digital beacons.
Mission Radio	Technisonic TDFM-136B (26df5da8...)	Tactical VHF-FM transceiver operating on 136–174 MHz. Used for air-to-ground tactical coordination with CAP ground teams and mission base.

3.2 The Critical GMA 1347 Audio Panel Trap

The Civil Air Patrol *Mission Observer Guide* emphasizes that nearly every reported in-flight radio failure is actually an audio panel configuration error. The GMA 1347 enforces strict separation between receiver audio and transmitter microphone selection:

- **Listening to COM3:** Pressing the **COM3** audio key lights the annunciator and routes VHF-FM radio audio to your headset. *This does NOT allow you to transmit on COM3!*
- **Transmitting on COM3:** You must press the **COM3 MIC** button on the top row. The illuminated microphone button dictates which transmitter receives your PTT signal.

CRITICAL COCKPIT TRAP — AUDIO VS. MIC SELECTION Selecting COM3 audio enables you to hear mission traffic, but keying the flight yoke or hand mic will transmit on COM1 (VHF Airband / ATC) if the COM1 MIC button remains selected. Always verify both the COM3 Audio and COM3 MIC annunciators are illuminated before making tactical transmissions.

3.3 Direction Finder: Becker SAR-DF 517 vs. RHOTHETA RT-600

A primary training objective of the CAP V-SAR simulator is demonstrating the operational coverage difference between the legacy Becker 517 and the modern fleet-standard RHOTHETA RT-600:



Figure 3: Dedicated C182T MO Station — Layout 2 (RHOTHETA RT-600 & TDFM-136B)

- **Becker SAR-DF 517 (Emergency Mode):** The 406 COSPAS/SARSAT receiver is locked

to a *single fixed frequency: 406.025 MHz*. If a modern beacon transmits on 406.037 MHz, the Becker in Emergency mode receives **zero signal**.

- **RHOTHETA RT-600 (CPSAR Groups):** The RT-600 receiver features seven broad-band CPSAR groups (A through S) covering 19 distinct 3 kHz channels. When set to **CPSAR2**, the RT-600 detects beacons across 406.031, 406.034, and 406.037 MHz simultaneously without retuning.
- **Becker Training Mode:** When switched to Training Mode, the Becker becomes freely tunable across 400–410 MHz, allowing the observer to manually tune to 406.037 MHz to re-acquire signal.

4. Instructor & Exercise Control Console

The Instructor Console (hosted on `http://ms01:8700/`) gives training officers comprehensive control over simulated flight exercises, real-time aircraft tracking, and dynamic environmental failure injection.

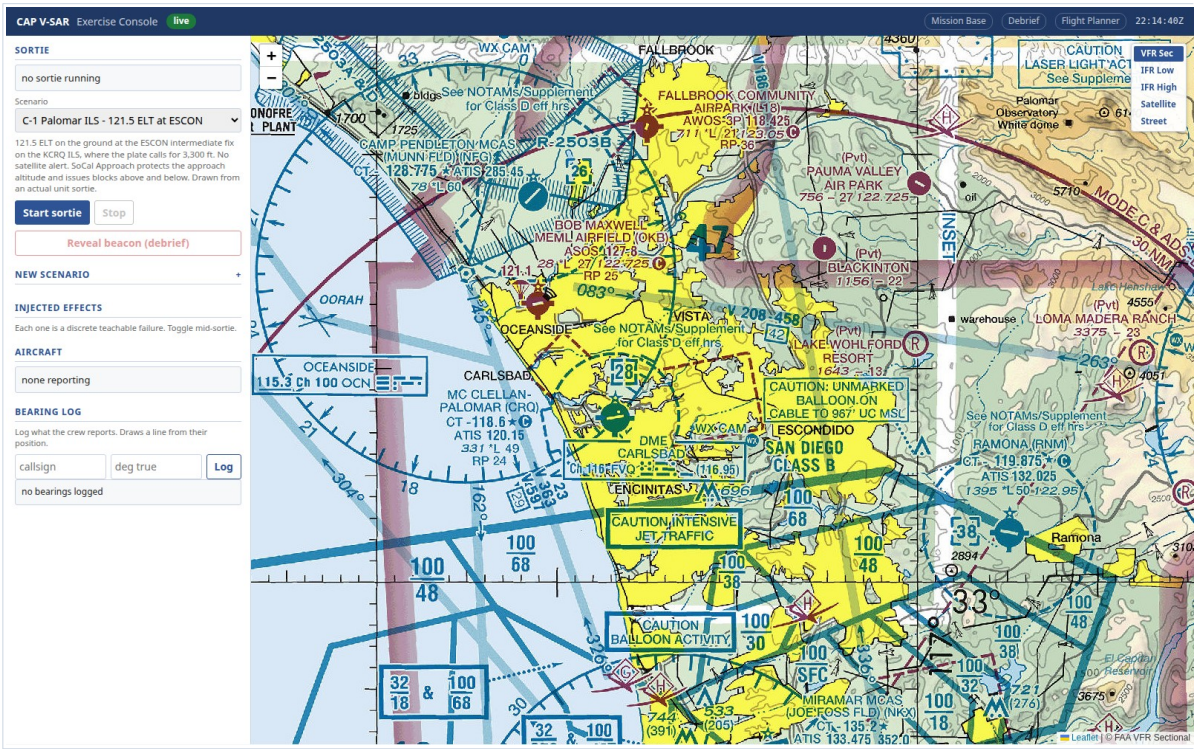


Figure 4: Instructor Control Console with Sectional Basemap & Dynamic Failure Controls

4.1 Scenario Selection & Interactive Creation

Instructors can activate pre-scripted training sorties (e.g., Palomar ILS ESCON ELT) or construct custom scenarios using the built-in **New Scenario** panel:

1. Click the **New scenario** accordion.
2. Click **Pick on map** and click anywhere on the FAA VFR Sectional chart to drop the beacon pin.
3. Drag the amber pin to fine-tune placement (e.g. into a narrow box canyon or behind a ridge).
4. Specify transmitter power (watts), ground elevation (ft MSL), and transmission band (121.5 / 243 / 406 MHz).
5. Click **Create scenario** — the new sortie is immediately registered and ready to fly.

4.2 Injected Propagation Effects

To build realistic electronic search skills, instructors can toggle discrete radio propagation effects during flight:

- **Radio Horizon (Line-of-Sight):** Computes spherical earth curvature and terrain shielding. The DF receiver remains silent until the crew climbs above the visual radio horizon.
- **406 Burst Cadence & Timehold:** Emulates true 406 MHz beacon behavior — an instantaneous 0.5-second burst every 50 seconds. The DF head holds the decoded bearing during the 50-second silence.
- **Cone of Confusion:** When the aircraft flies directly over the beacon, the relative bearing needle spins and the signal drops momentarily, confirming station passage.
- **Multipath Reflections:** Injects mountain-ridge reflections that deflect the apparent bearing by up to 40°, testing the crew's ability to cross-check bearings.

5. Mission Base & Debrief Modules

5.1 Mission Base Operator View

The Mission Base console (`http://ms01:8700/base`) represents the situational map for the Mission Radio Operator (MRO) and Air Operations Branch Director (AOBD). In accordance with CAP training standards, **beacon truth is hidden** on this view.



Figure 5: Mission Base Console with Aircraft Radar Track and Tactical Message Log

As the aircrew radios in "Ops Normal" reports and bearing lines, Mission Base operators log the cuts into the console. The system projects colored bearing rays across the sectional map, allowing the staff to visualize the intersection box and direct ground search teams.

5.2 Post-Flight Debrief Replay

After the sortie is terminated, the instructor and aircrew open the **Debrief Console** (`http://ms01:8700/debrief`). The debrief engine replays the recorded flight second-by-second against the unmasked beacon truth:

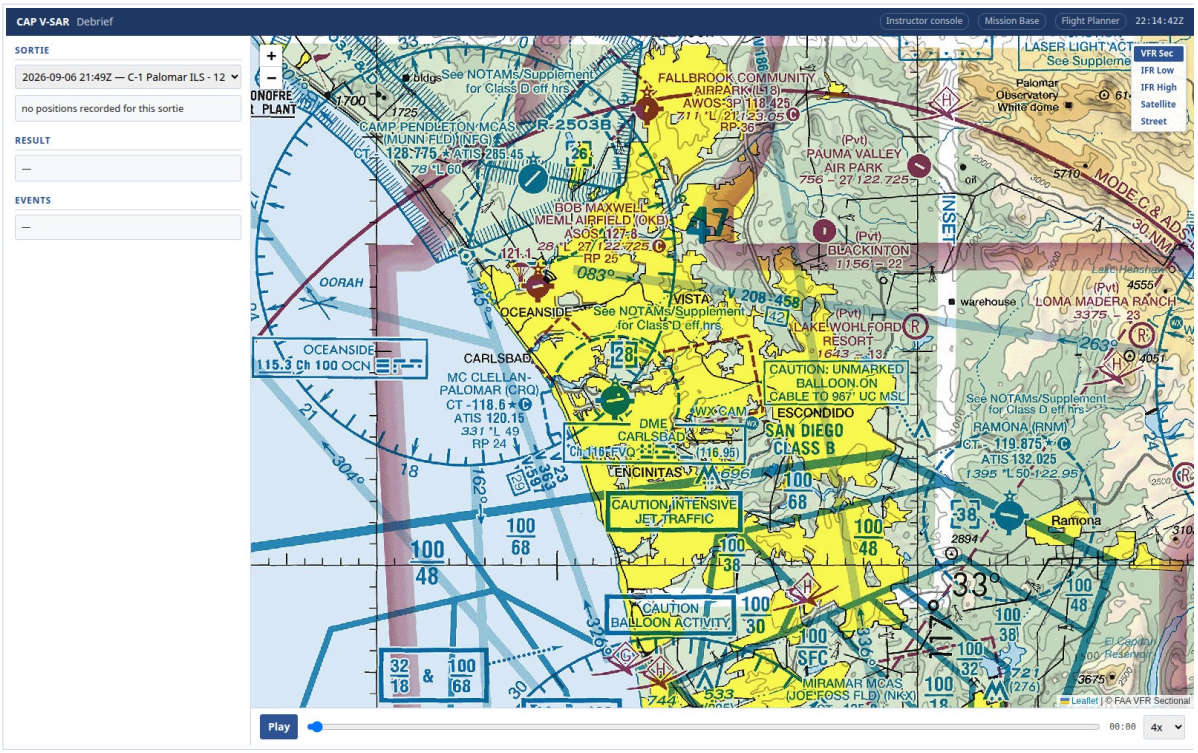


Figure 6: Debrief Console with Flight Track Scoring & Bearing Error Analysis

- **Closest Point of Approach (CPA):** Exact distance in nautical miles and timestamp of closest approach.
- **Track Efficiency:** Compares total miles flown against direct-line distance to evaluate pattern discipline.
- **Bearing Cut Accuracy:** Displays reported cockpit bearings against truth vectors to demonstrate multi-path errors and averaging lag.

5.3 Mission Radio Operator (MRO) Base Station Radio Console

De-coupled from the situational radar map, the **Base Radio Console** (`http://ms01:8700/radio`) provides a standalone, touch- and mouse-operable communications station for Mission Radio Operators using tablets (e.g. iPad) or laptops. The application emulates the two primary ground radio systems deployed across Civil Air Patrol base stations:

Motorola APX 6500 (VHF-FM Tactical & Land Mobile Net)

The standard CAP tactical mobile and base transceiver, providing digital P25 and analog FM communications across 136–174 MHz. Used by the MRO to manage the primary mission net, coordinate ground search teams, and maintain air-to-ground tactical contact with the Mission

Observer on the airborne Technisonic TDFM-136B radio:

- **Zone 1 (Tactical Net):** `TAC 1` (148.150 MHz), `TAC 2` (148.125 MHz), `CC 1` (150.225 MHz P25 NAC $293), `CC 2` (149.275 MHz P25), `AIR 1` (150.5375 MHz), `AIR 2`, `SAR 1`, and `GUARD`.
- **Physical Controls & Display:** High-contrast vacuum-fluorescent green-cyan OLED display with RSSI signal meter; rotary VOL/PWR and CHANNEL knobs with touch steppers; illuminated softkeys (`HOME`, `ZONE ▲/▼`, `SCAN`, `MON`, `DIM`); and tri-color TX/RX/SCAN LEDs.
- **Acoustic Simulation:** Authentic Motorola 3-tone P25 talkgroup permit chirp (941/1336/1209 Hz) upon keying PTT, followed by digital squelch tail on release.

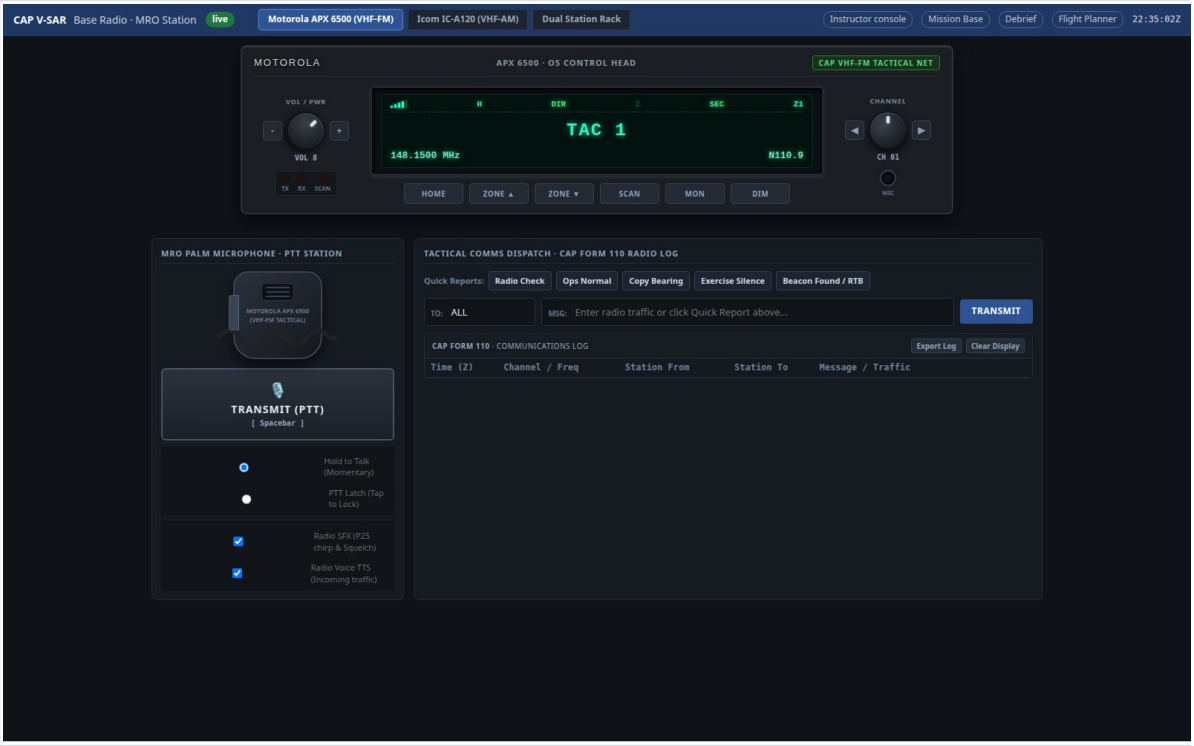


Figure 7: Motorola APX 6500 VHF-FM Tactical Base Radio & MRO Dispatch Console

Icom IC-A120 (VHF-AM Aeronautical Transceiver)

The standard desktop and vehicle aviation transceiver deployed at CAP base stations, staging hangars, and flight line dispatch to communicate directly with search aircraft on aeronautical VHF-AM frequencies (118.000–136.975 MHz):

- **Channel Presets:** Pre-programmed CAP aeronautical channels: `CH 01` (123.100 MHz — SAR Primary), `CH 02` (122.900 MHz — SAR Secondary / Multicom), `CH 03` (121.500 MHz

— Emergency Guard / ELT Monitor), `CH 04` (122.775 MHz — Air-to-Air), `CH 05` (123.050 MHz — Unicom), and `CH 06` (126.200 MHz — Military SAR).

- **VFO Tuning & Display:** Ice-blue backlit LCD with large 7-segment frequency digits, memory alias, audio bar meter, and rotary tuning dial with 25 kHz / 1 MHz push-toggle stepping.
- **Acoustic Simulation:** Authentic VHF-AM mechanical relay click, 1000 Hz sidetone hiss, and squelch noise burst (*shhh-klick*).



Figure 8: Icom IC-A120 VHF-AM Aeronautical Air-to-Ground Base Station Console

Dual Station Equipment Rack & CAP Form 110 Comms Log

The console supports a **Dual Station Rack View** (`/radio?mode=dual`) running both radios concurrently for dual watch. The dispatch section includes a live **CAP Form 110 Communications Log** that automatically records all transmissions with UTC timestamps and network tags, one-touch tactical canned reports, and a plain-text log export tool.

6. Dedicated Civil Air Patrol Cessna 182T NAV III (G1000 CAP) Aircraft

To fulfill the operational need for an authentic, high-performance search platform in X-Plane 12, a dedicated **Cessna 182T Skylane NAV III** (tail number `N499CP`) has been engineered as a standalone airframe package. Adapted from Laminar Research's high-fidelity Cessna 172SP G1000 base model via `scripts/build_c182t_package.py`, the package incorporates complete Lycoming IO-540 flight dynamics, a 3-blade constant-speed propeller, and automatic Garmin G1000 Manifold Pressure EIS instrumentation.



Figure 9: Civil Air Patrol Cessna 182T NAV III (N499CP) Aircraft Package & Selection Banner

Powerplant & Propeller Integration

- **Lycoming IO-540-AB1A5 Engine:** 230 HP @ 2,400 RPM (541 cu in, fuel-injected, direct-drive).
- **McCauley 3-Blade Constant-Speed Propeller:** Modeled with governor blade pitch ranges from 10° to 40°.
- **G1000 EIS Manifold Pressure Auto-Activation:** In X-Plane 12, configuring a constant-speed propeller (`_prop_type = 1`) automatically instructs the Garmin G1000 MFD to render the **Manifold Pressure (`MAN IN HG`)** gauge alongside engine **RPM**, presenting

identical indications to the real C182T NAV III cockpit.

- **Cowl Flaps:** Manual cowl flap system modeled for thermal and aerodynamic climb management.

Airframe Specifications & Performance Envelope

Maximum Takeoff Weight (MTOW)	3,100 lbs (1,406 kg)	Certified gross weight for full 3-person crew + mission gear
Basic Empty Weight	1,970 lbs (894 kg)	Useful load of 1,130 lbs
Total Usable Fuel	87 gal (522 lbs / 100LL)	5.5+ hour search endurance at 65% power loiter
Stall Speed — Clean (V_S)	51 KIAS	Flaps up, gross weight stall threshold
Stall Speed — Flaps Full (V_{S0})	41 KIAS	Slow-flight search loiter safe margin
Best Angle of Climb (V_X)	65 KIAS	Short-field and mountain ridge clearance
Best Rate of Climb (V_Y)	80 KIAS	Enroute altitude attainment (approx 924 fpm @ sea level)
Best Glide Speed (V_{BG})	76 KIAS	Engine-out maximum gliding range
Flap Extension Speeds (V_{FE})	140 KIAS (10°) / 100 KIAS (20°–30°)	Approach and pattern deceleration gate
Max Structural Cruising (V_{NO})	140 KIAS	Smooth-air search transit speed
Never Exceed Speed (V_{NE})	175 KIAS	Airframe structural dive limitation

7. Training Progression: Rehearsing a SAREX Before Flying It

A Search and Rescue Exercise (SAREX) is Civil Air Patrol's structured training event: a mission

base under an Incident Commander (IC) tasks aircrews and ground teams against a scripted incident. CAP V-SAR is designed so that the SAREX is the **second** time a crew flies its missions, not the first. The exercise director writes the incident once; the wing runs it three times on the simulator — two weeks out, one week out and two days out — and then flies it for real. The live exercise reinforces skills learned at home instead of introducing them.

7.1 Writing the Incident Once

The scripted incident is built the way a real IC builds ICS 204 assignments. Example: a rented C182 is overdue on a VFR flight from Carlsbad (KCRQ) to Thermal (KTRM); its registered 406 MHz ELT has produced a COSPAS-SARSAT composite position in the mountains east of Julian. Three air tasks fall out of that:

A	Route search	Creeping line along the intended route, 1 nm spacing	Visual search for the aircraft along its planned track
B	ELT search	Transit to the composite; home on the 406 burst and 121.5 signal; expanding square from station passage	A position good enough to hand to a ground team
C	Grid search	Parallel track of a 15-minute grid quadrant, 1 nm spacing	Cover the case where the ELT position is wrong

The instructor creates the scenario in the Exercise Control Console (§4): beacon on a ridge, 406.025 MHz training frequency, and an initial effect set of *radio horizon* and *terrain masking* only. Beacon truth is withheld from every other view until debrief.

7.2 The Three Sessions

1 — T-14 days Mission base stands up	Nothing is flown. The AOBD builds the three patterns in the Flight Planner from the ICS 204s, exports each as an X-Plane `.fms` and pushes it to the base map; the terrain profile catches a leg with poor clearance. The MRO joins the mission net and works the channel	Every seat's equipment and the base's communications rhythm are familiar before the mission starts.

	plan. The MOs drill the MO station: GMA 1347 COM3 MIC selection (§3.2), TDFM-136B channel entry, DF head power-up, squelch, scan, and tuning 406.025 not 406.037. Radio checks and the first 30-minute ops-normal cycle.	
2 — T-7 days Full mission	The IC briefs on Mumble. Sortie A flies the creeping line and calls each leg; Sortie C flies the grid and meets *wing blanking* in a turn. Sortie B homes on the 406 burst; the instructor injects *cone of confusion* as the aircraft nears the beacon, then *multipath*. The MO takes one bad bearing; the AOBD's plotted cut disagrees with the previous three and the base asks the crew to re-fly the leg. The IC retasks Sortie A onto a second grid through the MRO. Debrief reveals the beacon and scores every bearing against truth.	Recognising station passage and multipath; base staff cross-checking the cockpit; retasking and ops-normal discipline with three aircraft airborne. The SET notes which SQTR tasks were demonstrated.
3 — T-2 days Dress rehearsal	Everything is now exactly as it will be on the live day: real callsigns, the real communications plan, the real grid numbers and timing, the same base staff in the same roles, and the composite position moved to where the practice beacon will actually be. Crews fly it cold; the instructor injects the effects that gave trouble in Session 2.	Confirmation. Sortie B finds the beacon on the first expanding square; the bearing plot has no rejected cuts; ops-normal reports are on time for all three aircraft.
Live SAREX	The same three assignments against a real practice beacon on 121.775 MHz, with real terrain, weather, radios and a ground team.	Every procedure has been performed at least twice; the live day adds the airplane, the weather and the pressure. The SET signs off tasks demonstrated on both the rehearsal and the live sortie.

7.3 What the System Measures Across the Sessions

Bearing error per report (degrees)	Debrief Console	DF technique; multipath and cone

	(§5.2)	recognition
Closest point of approach; time to first good cut	Debrief Console	Homing effectiveness
Track efficiency against the assigned pattern	Debrief Console	Pattern discipline
Ops-normal report timing per aircraft	Form 110 / message log (§5.3)	Communications discipline at base and in the cockpit
Rejected or corrected cuts	Base bearing log (§5.1)	Whether base staff cross-check the cockpit
Session-to-session change in all of the above	Three debriefs	Whether the rehearsal worked

The full narrative, including the current system limitations that apply to a three-aircraft rehearsal, is in `docs/USE_CASE_SAREX_REHEARSAL.md`.

8. Quick Reference Checklists (QRC)

Mission Radio Operator (MRO) Base Station Checklist

1. MRO Console URL	Open `http://ms01:8700/radio` on tablet or dispatch laptop
2. Radio Mode Selection	Select **Motorola APX 6500** (FM Tactical), **Icom IC-A120** (AM Aviation), or **Dual Station Rack**
3. Power & Volume Check	Turn VOL knob clockwise; verify display illuminates and audio sidetone is active
4. APX 6500 Tactical Net	Verify Zone 1, select `TAC 1` (148.150 MHz) or assigned mission net
5. IC-A120 Airband Net	Verify `CH 01` (123.100 MHz SAR PRI) or assigned air-to-ground frequency

6. Squelch Adjustment	Adjust SQL knob until background receiver hiss cuts; verify SQL flag clears
7. Comms Check with Sortie	Click **Radio Check** or key PTT (Spacebar): *"CAP499, Mission Base: Radio check, how do you copy?"*
8. CAP Form 110 Logging	Confirm transmission appears in Form 110 log; export text log upon sortie termination

Mission Observer (MO) Cockpit Pre-Flight Checklist

1. Air Manager Panel	Launch "Cessna 182T MO Station", select Becker 517 or RT-600 layout
2. G1000 MFD Display	Verify MFD function, map orientation North-Up, 360-degree Arc
3. GMA 1347 Audio Panel	COM3 Audio ON (green LED), ICS Intercom VOX set, volume comfortable
4. GMA 1347 Mic Selector	COM3 MIC selected for tactical ground team comms; COM1 for ATC
5. TDFM-136B Power / Channel	Power knob ON, select mission channel, squelch open check
6. Direction Finder Power	PAGE knob clockwise from OFF to Page 1, display illuminates
7. DF Band & Frequency	Set 121.500 MHz (VHF) or 406.025 / CPSAR2 (COSPAS/SARSAT)
8. DF Squelch Setting	Rotate SQUELCH knob until audio hiss cuts, verify SQL annunciator

In-Flight Electronic Search & Homing Flow

1. Initial Detection	Notice DF audio sweep tone; verify signal level bars > 0
2. Bearing Confirmation	Observe relative bearing pointer; confirm spread is under 20 degrees

3. Pilot Coordination	Call heading correction to pilot: *"Observer to Pilot: steer right 020 to home"*
4. 406 MHz Burst Hold	Note timehold indicator; maintain heading between 50-second bursts
5. Station Passage (Cone)	Watch for bearing needle spin and rapid signal dip; call *"Station passage, over target"*
6. Mission Base Report	Key COM3 MIC on TDFM-136B: *"Mission Base, CAP499 has target located at..."*